HP_power_board

File: HP_power_board.kicad_sch

LP_logic_board

File: LP_logic_board.kicad_sch

pi_power_stub

File: pi_power_stub.kicad_sch

pack_sense_board

File: pack_sense_board.kicad_sch

marengo

Sheet: /

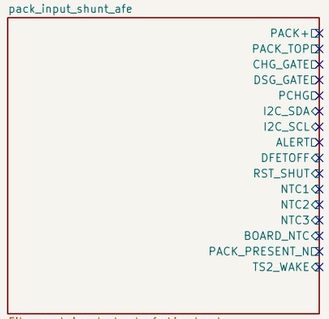
File: marengo_pdu.kicad_sch

Title: marengo power board rev A

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 1/27

Sheet pins carry no-connect flags until the FET, E-stop and LP sheets exist.

pack_input_shunt_afe



File: pack_input_shunt_afe.kicad_sch

bms_fets_precharge

File: bms_fets_precharge.kicad_sch

estop_banks

File: estop_banks.kicad_sch

vmot_bulk_clamp

File: vmot_bulk_clamp.kicad_sch

port_channel_1

File: port_channel.kicad_sch

port_channel_2

File: port_channel.kicad_sch

port_channel_3

File: port_channel.kicad_sch

port_channel_4

File: port_channel.kicad_sch

port_channel_5

File: port_channel.kicad_sch

port_channel_6

File: port_channel.kicad_sch

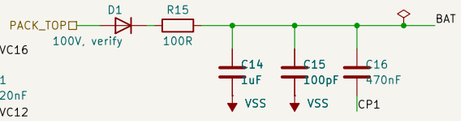
marengo

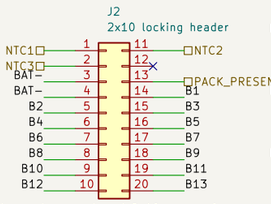
Sheet: /HP_power_board/
File: HP_power_board.kicad_sch

Title: HP_power_board: high-power board

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 2/27

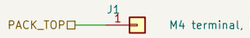
BAT from the pack + terminal, so the AFE stays up if J2 is pulled
with pack power still connected. Fig 16-2: diode, 100R, 1uF, 100pF, 0.47uF CP1. Caps 100 V.

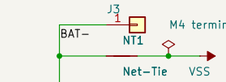


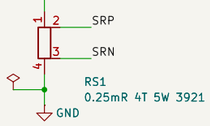


J2: harness to the pack sense board 2x10, same pin map.
Columns 4-10 zigzag B0..B13: a bent pin shorts 1 cell
across a column, 2 cells along a row.
NTC return = B0 at the pack sense board. Spare: pin 12.
B0 lands on BAT− (datasheet p70: VC0 tied to VSS on PCB).

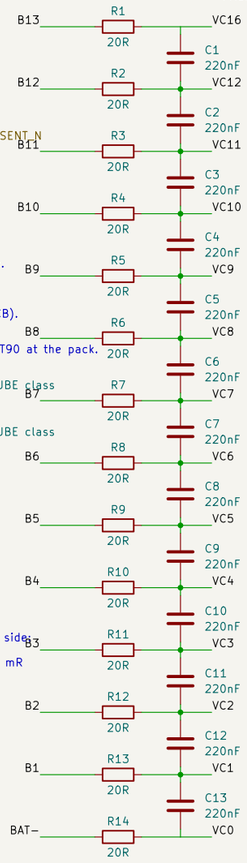
J1 pack +, J3 pack −: 10 AWG ring-lug pigtails to the XT90 at the pack.





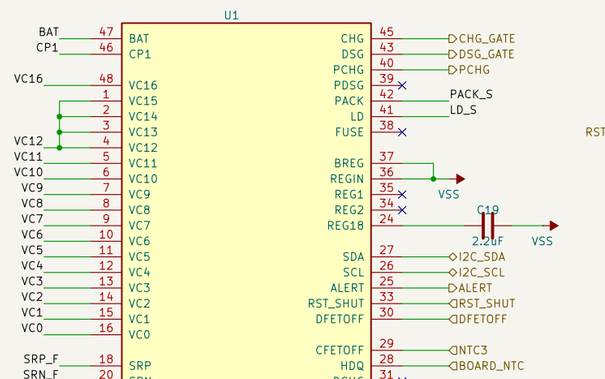


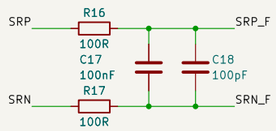
Pack negative -> shunt -> system GND. SRP is the cell side;
charge current makes SRP positive (datasheet p5).
AFE VSS ties at BAT−, so VSS sits above GND by I x 0.25 mR
in discharge: 40 mV at the 160 A SCD point.



13S per Table 16-3 (p75): cells 1, 2, 16 must be real.
VC0..VC12 = cells 1-12, VC15-VC16 = cell 13.
VC13-15 short to VC12. Cell inputs in use: 1-12 and 16.

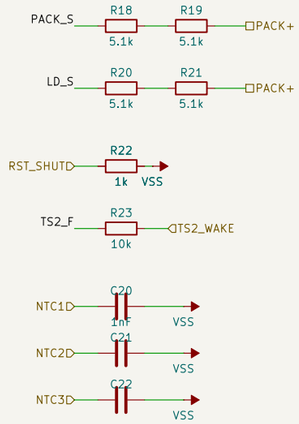
Cell filter: 20R + 220nF on every used input
(datasheet p9: RC 20-100R, CC 0.1/0.22/1 uF; p67 Fig 16-2).





p65, p76: 100R each side, 100nF + 100pF differential, near U1,
symmetric, same PCB side. Kelvin SRP/SRN routed in parallel from RS1.

PACK and LD both sense the common port after the FETs
(Fig 16-3 R42-R45). PDSG, FUSE, DCHG, DDSG unused (p75).



Four thermistors: TS1, TS3, CFETOFF (NTC1-3), HDQ (board).
TS2 stays free as SHUTDOWN wake (p22 pin table, section 13):
switch to VSS through 10k (Fig 16-2 R27). 1nF: p65 limit <4nF.
REGIN, BREG tied to VSS, REG1, REG2 open (p75): I2C pull-ups on LP.
CFETOFF as NTC needs DFETOFF set as BOTHOFF to stop CHG by pin (p48).

SIZING (design/README 2.2, shunt 0.25 mOhm; datasheet p22 steps)
OCD1 16 mV = 64 A (hardware comparator, up to 425 ms) OCD3 60 A for 1 s (current-based, inside the AFE, the agreed 60 A ceiling)
OCC 12 mV = 48 A at 425 ms SCD 40 mV = 160 A, +/-15% = 136-184 A, delay 15-450 us in 15 us steps
OCD and OCC are set in 2 mV steps. Accuracy below 20 mV is -2/+2.65 mV (p22), about -8/+10.6 A on this shunt.

OPEN ON THIS SHEET
1. PACK_PRESENT_N: loop pulled low through BAT- when J2 is mated. PLAN says it opens DSG directly; the logic joining it to DFETOFF is not captured here.
2. External balance FETs (PLAN 3.1, 150 mA per cell) are not captured. They change the cell RC network: revisit 20R.
3. CP1 0.47 uF is the datasheet value. Check against the gate charge of the 4+4 FET banks (p65 example: 2.2 uF for 400 nF of gate load).
4. I2C pull-ups, ALERT pull-up and the DFETOFF/CFETOFF polarity go with the LP sheets and OTP settings.
5. No footprints yet for J1, J3 (M4 terminal part not chosen) or RS1 (no 3921 4-terminal in the library). J2 footprint assumes Micro-Fit vertical.

Pack terminals, sense header, shunt, BQ76952 AFE. Part numbers verify. TI SLUSE13B.

marengo

Sheet: /HP_power_board/pack_input_shunt_afe/
File: pack_input_shunt_afe.kicad_sch

Title: pack_input_shunt_afe

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 3/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/bms_fets_precharge/
File: bms_fets_precharge.kicad_sch

Title: bms_fets_precharge: BMS CHG/DSG banks, precharge

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 4/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/estop_banks/
File: estop_banks.kicad_sch

Title: estop_banks: E-stop banks A and B

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 5/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/vmot_bulk_clamp/
File: vmot_bulk_clamp.kicad_sch

Title: vmot_bulk_clamp: VMOT bulk, TVS, regen clamp

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 6/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/port_channel_1/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 7/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/port_channel_2/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 8/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/port_channel_3/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 9/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/port_channel_4/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 10/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/port_channel_5/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 11/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /HP_power_board/port_channel_6/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 12/27

supervisor_mcu

File: supervisor_mcu.kicad_sch

can_bridges_usb_hub

File: can_bridges_usb_hub.kicad_sch

isolated_can_1

File: isolated_can.kicad_sch

isolated_can_2

File: isolated_can.kicad_sch

isolated_can_3

File: isolated_can.kicad_sch

isolated_can_4

File: isolated_can.kicad_sch

isolated_can_5

File: isolated_can.kicad_sch

isolated_can_6

File: isolated_can.kicad_sch

front_end_12v_24v

File: front_end_12v_24v.kicad_sch

pol_rails

File: pol_rails.kicad_sch

rail_switches_meters

File: rail_switches_meters.kicad_sch

io_fans_estop_loop

File: io_fans_estop_loop.kicad_sch

marengo

Sheet: /LP_logic_board/
File: LP_logic_board.kicad_sch

Title: LP_logic_board: low-power board

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 13/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/supervisor_mcu/
File: supervisor_mcu.kicad_sch

Title: supervisor_mcu

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 14/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/can_bridges_usb_hub/
File: can_bridges_usb_hub.kicad_sch

Title: can_bridges_usb_hub

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 15/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/isolated_can_1/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 16/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/isolated_can_2/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 17/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/isolated_can_3/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 18/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/isolated_can_4/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 19/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/isolated_can_5/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 20/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/isolated_can_6/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 21/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/front_end_12v_24v/
File: front_end_12v_24v.kicad_sch

Title: front_end_12v_24v

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 22/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/pol_rails/
File: pol_rails.kicad_sch

Title: pol_rails

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 23/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/rail_switches_meters/
File: rail_switches_meters.kicad_sch

Title: rail_switches_meters

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 24/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /LP_logic_board/io_fans_estop_loop/
File: io_fans_estop_loop.kicad_sch

Title: io_fans_estop_loop

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 25/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /pi_power_stub/
File: pi_power_stub.kicad_sch

Title: pi_power_stub

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 26/27

Placeholder. Not yet captured.

Placeholder, not yet captured

marengo

Sheet: /pack_sense_board/
File: pack_sense_board.kicad_sch

Title: pack_sense_board

Size: A4	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 27/27